

Prime Quintuplets and the Sextuplet Boundary for $Q(n) = n^{47} - (n-1)^{47}$

Extreme Prime Clusters in a High-Degree Polynomial

Version 2 (revised and corrected)

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Abstract

Among the 742 prime quadruplets found for $Q(n) = n^{47} - (n-1)^{47}$ over $1 \leq n \leq 2 \times 10^{11}$ [6], exactly **7 extend to prime quintuplets** (five consecutive integers all generating probable primes of 488–520 digits), while **no sextuplet** was found. This paper develops a quantitative theory for this emergence–extinction transition.

$Q(n)$ is the homogenised 47th cyclotomic polynomial $\Phi_{47}(n, n-1)$. Consequently every prime divisor of every value $Q(n)$ is congruent to 1 (mod 47), and the root count is *bifurcated*: $\omega_1(p) = 0$ for $p \not\equiv 1 \pmod{47}$ and $\omega_1(p) = 46$ for the *resonant primes* $p \equiv 1 \pmod{47}$. The resonant primes impose a periodic sieve obstruction that produces “cliffs” in the Bateman–Horn Euler product, the first at $p = 283$. Because the plain Euler product converges only conditionally (it still oscillates by $\pm 20\%$ for sieve bounds up to 10^4), we evaluate the singular series *exactly* by factoring out the Dirichlet L -values $\prod_{\chi \neq \chi_0} L(1, \chi) = 0.1151693$ modulo 47, which leaves an absolutely convergent product. This gives $\mathfrak{S}_4 = 6,514$, $\mathfrak{S}_5 = 58,550$ and $\mathfrak{S}_6 = 535,500$ with no calibration to the data. The resulting parameter-free predictions, $E[C_4] = 747$, $E[C_5] = 5.9$ and $E[C_6] = 0.047$ at $N = 2 \times 10^{11}$, agree with the observed 749 quadruplet positions, 7 quintuplets and 0 sextuplets; in the pioneer zone ($N = 2 \times 10^9$) the predicted numbers of primes and prime pairs agree with the census to 0.05% and 0.07%. The first sextuplet is predicted at $N^* \approx 1.0 \times 10^{13}$, a ~ 50 -fold extension of the present search, where the primes have ~ 601 digits. The inter-level suppression factor is not a constant: it equals $(I_k/I_{k+1})/(\mathfrak{S}_{k+1}/\mathfrak{S}_k) \approx 46 \ln N/9$, i.e. ≈ 103 at $N = 2 \times 10^9$ and ≈ 127 at $N = 2 \times 10^{11}$, which reconciles the pioneer-zone ratios with the full-range one.

Keywords: prime quintuplets, sextuplet boundary, Bateman–Horn conjecture, singular series, cyclotomic polynomial, resonant primes, Dirichlet L -functions, constellation hierarchy, high-degree polynomials

MSC 2020: 11N32, 11A41, 11Y11, 11Y16

Changes in Version 2. Version 1 [7] tabulated partial Euler products that, except at the sieve bound $B = 10^4$, were not the output of the published script; the convergence claim based on a “calibration factor of 0.999” was therefore unsupported. Version 2 replaces that table with the correct partial products (Table 2), adds an exact L -function evaluation of the

singular series (Section 3.2), removes the empirical calibration, corrects the digit counts of the quintuplets (each was one too small), corrects the Kolmogorov–Smirnov statistic, the prime count in the “unobstructed corridor”, the Poisson interval, the cost estimate, and replaces the claim of a constant suppression factor by the correct $\ln N$ dependence. A complete list is given in `CHANGELOG_v2.md` in the data repository.

1 Introduction

A *prime k -constellation* of the polynomial $Q(n) = n^{47} - (n-1)^{47}$ is a run of k consecutive integers $n, n+1, \dots, n+k-1$ such that all k values of Q are (probable) primes. The census of the pioneer zone ($n \leq 2 \times 10^9$) [5] established a strict morphological hierarchy—18.1 million solitary primes, 173,351 pairs, 1,749 triplets, 15 quadruplets—with inter-level ratios of approximately $100\times$. The deep-space census [6] extended the quadruplet search to $N = 2 \times 10^{11}$, finding 742 quadruplets, 7 of which are quintuplets, and zero sextuplets.

The present paper (Part III of the Titan Project) addresses the central question arising from these data:

Why do exactly 7 quintuplets emerge from 742 quadruplets, and why does the sextuplet order remain vacant across 200 billion integers?

We answer it through:

- (i) The identification of $Q(n)$ with the homogenised cyclotomic polynomial $\Phi_{47}(n, n-1)$, which gives the *bifurcated root structure*: zero roots for most primes, but exactly 46 roots modulo every prime $p \equiv 1 \pmod{47}$ (the *resonant primes*);
- (ii) An *exact* evaluation of the singular series $\mathfrak{S}_1, \dots, \mathfrak{S}_6$ by factoring out the Dirichlet L -values modulo 47, together with the (slowly, conditionally convergent) plain Euler products for comparison;
- (iii) A parameter-free Bateman–Horn validation across the whole hierarchy, from 18 million primes down to 7 quintuplets;
- (iv) A quantitative prediction for the sextuplet boundary at $N^* \approx 1.0 \times 10^{13}$;
- (v) Analysis of the spatial distribution of the 7 quintuplets and of the N -dependence of the inter-level suppression factor.

1.1 Counting conventions

The Bateman–Horn conjecture [1] predicts the number of *positions* $n \leq N$ such that the k values $Q(n), \dots, Q(n+k-1)$ are all prime:

$$C_k(N) \sim \mathfrak{S}_k I_k(N), \quad I_k(N) = \int_2^N \frac{dt}{\prod_{i=0}^{k-1} \ln Q(t+i)} \approx \int_2^N \frac{dt}{(46 \ln t + \ln 47)^k}, \quad (1)$$

since $Q(t) = 47t^{46} - \binom{47}{2}t^{45} + \dots$. Version 1 used $(46 \ln t)^k$ in the denominator; the $\ln 47$ term lowers I_4 by 1.3% and I_6 by 2.0% at $N = 2 \times 10^{11}$ and is kept throughout this version. A census, on the other hand, reports *maximal runs*: a quintuplet is listed once, not also as two quadruplets. A maximal run of length m contains $m-k+1$ positions of order k , so the 742 quadruplets of Part II (735 of exact length 4 plus the 7 quintuplets) correspond to $735 + 2 \times 7 = 749$ quadruplet positions. All comparisons with (1) below use position counts.

2 The Bifurcated Root Structure of $Q(n)$

2.1 $Q(n)$ as a cyclotomic value

Since $x^{47} - 1 = (x - 1)\Phi_{47}(x)$ with $\Phi_{47}(x) = x^{46} + \cdots + 1$,

$$Q(n) = n^{47} - (n-1)^{47} = (n - (n-1))(n-1)^{46} \Phi_{47}\left(\frac{n}{n-1}\right) = \Phi_{47}(n, n-1), \quad (2)$$

the homogenised 47th cyclotomic polynomial evaluated at the coprime pair $(n, n-1)$. It is irreducible of degree 46 with leading coefficient 47. A classical theorem on the divisors of $a^q - b^q$ [3, 4] states that every prime divisor of $\Phi_q(a, b)$ with $\gcd(a, b) = 1$ and q prime is either q itself or congruent to 1 (mod q). Because $n^{47} \equiv n \pmod{47}$ by Fermat, $Q(n) \equiv n - (n-1) = 1 \pmod{47}$, so 47 never divides $Q(n)$. Hence:

Proposition 2.1. *Every prime factor of every value $Q(n)$, $n \geq 2$, is congruent to 1 (mod 47). In particular $Q(n)$ is odd, prime to 3, and prime to every prime < 283 .*

2.2 Main Theorem

Theorem 2.2 (Bifurcated Root Structure). *Let p be a prime and $Q(n) = n^{47} - (n-1)^{47}$. Then:*

- (a) *If $p \nmid n(n-1)$, then $Q(n) \equiv 0 \pmod{p}$ if and only if $n/(n-1)$ is a nontrivial 47th root of unity modulo p . Such roots exist if and only if $p \equiv 1 \pmod{47}$.*
- (b) *If $p \mid n$ or $p \mid (n-1)$, then $Q(n) \equiv 1 \pmod{p}$.*
- (c) **Root count:**

$$\omega_1(p) = \begin{cases} 46 & \text{if } p \equiv 1 \pmod{47}, \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

Proof. (a) Suppose $p \nmid n(n-1)$ and write $a = n$, $b = n-1$. Then $Q(n) \equiv 0$ iff $(a/b)^{47} \equiv 1 \pmod{p}$. Let $g = \gcd(47, p-1) \in \{1, 47\}$. If $g = 1$ the only solution of $x^{47} = 1$ in $\mathbb{F}_p^*$ is $x = 1$, forcing $a \equiv b$, i.e. $1 \equiv 0$ —impossible; so $\omega_1(p) = 0$. If $g = 47$ then $p \equiv 1 \pmod{47}$ and $\mathbb{F}_p^*$ contains a cyclic subgroup of order 47 generated by $\zeta = g_0^{(p-1)/47}$, g_0 a primitive root. For each $j \in \{1, \dots, 46\}$ the equation $a/b = \zeta^j$ with $a = b + 1$ gives $b(\zeta^j - 1) = 1$, hence the unique root

$$n_j = 1 + (\zeta^j - 1)^{-1} \pmod{p}, \quad j = 1, \dots, 46, \quad (4)$$

and these 46 residues are distinct because $j \mapsto \zeta^j$ is injective. Both b and $a = \zeta^j b$ are units, so no root is lost to the excluded cases. (b) If $p \mid n$ then $Q(n) \equiv -(-1)^{47} = 1$; if $p \mid (n-1)$ then $Q(n) \equiv 1^{47} - 0 = 1$. $\square$

Theorem 2.2 is the statement that Φ_{47} splits completely modulo p exactly when p splits completely in $\mathbb{Q}(\zeta_{47})$, i.e. when $p \equiv 1 \pmod{47}$; since Q has degree 46, resonant primes carry the maximal possible number of roots.

Explicit example. The smallest resonant prime is $p = 283 = 6 \times 47 + 1$. A primitive 47th root of unity is $\zeta = 2^6 = 64$ (indeed $64 \not\equiv 1$ and $64^{47} \equiv 1 \pmod{283}$). For $j = 1$: $b = 63^{-1} \equiv 9 \pmod{283}$ since $63 \times 9 = 567 = 2 \times 283 + 1$; thus $n_1 = 10$ and $Q(10) = 10^{47} - 9^{47} \equiv 0 \pmod{283}$. Direct enumeration confirms exactly 46 roots modulo 283.

2.3 Resonant Primes and the Periodic Sieve Obstruction

Definition 2.3 (Resonant Primes). *A prime p is resonant for Q if $p \equiv 1 \pmod{47}$: $\mathcal{R} = \{283, 659, 941, 1129, 1223, 1693, \dots\}$. By Dirichlet's theorem a proportion $1/\varphi(47) = 1/46$ of all primes is resonant; there are 28 resonant primes below 10^4 (Dirichlet expectation 26.7) and 1,700 below 10^6 (expectation 1,706).*

For k -constellations the relevant quantity is $\omega_k(p)$, the number of residues r modulo p for which at least one of $Q(r), \dots, Q(r+k-1)$ vanishes. By (4) this is the size of the union of the k shifted root sets $\{n_j - i\}$, $0 \leq i < k$. Two shifted sets overlap only if $n_j - n_{j'} \equiv d \pmod{p}$ for some $d \in \{\pm 1, \dots, \pm(k-1)\}$, which happens only for primes dividing the norm of the algebraic number $(\zeta^j - 1)^{-1} - (\zeta^{j'} - 1)^{-1} - d \in \mathbb{Q}(\zeta_{47})$ —a finite set of primes for each k . Hence

$$\omega_k(p) = 46k \quad \text{for all but finitely many resonant } p, \quad \omega_k(p) = 0 \quad \text{for non-resonant } p. \quad (5)$$

Below 10^4 the exceptional resonant primes for $k = 6$ are 283, 659, 941, 1129, 1223, 1693, 1787, 2069, 2351, 2539, 2633, 3761, 4513, 4889, 5077, 5171, 6863 and 8461; from $p = 5641$ on, all but 6863 and 8461 attain $\omega_6 = 276$. Table 1 lists the first five resonant primes.

Table 1: Root counts $\omega_k(p)$ at the first five resonant primes and the local Bateman–Horn factors $\sigma_k(p) = (1 - \omega_k/p)/(1 - 1/p)^k$. For non-resonant primes $\omega_k = 0$ and $\sigma_k(p) = (p/(p-1))^k$.

p	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	$\sigma_4(p)$	$\sigma_5(p)$	$\sigma_6(p)$
283	46	78	102	126	148	164	0.563	0.486	0.430
659	46	88	124	160	194	228	0.762	0.711	0.660
941	46	92	134	176	216	256	0.816	0.775	0.733
1129	46	88	128	166	202	235	0.856	0.825	0.796
1223	46	88	125	162	195	226	0.870	0.844	0.819
generic $p \in \mathcal{R}$	46	92	138	184	230	276	$\approx 1 - 45k/p$		

The first resonant prime $p = 283$ is the most destructive: $\sigma_4(283) = 0.563$ nearly halves the running Euler product for quadruplets (Figure 1). Subsequent resonant primes produce progressively milder corrections as $\omega_k/p \rightarrow 0$. The 60 primes below 283 all have $\omega_k = 0$ (Proposition 2.1); this “unobstructed corridor” lets the product climb to $\mathfrak{S}_4(282) = 10,780$ before the first cliff and is the reason for the large absolute size of the constants.

3 Computing the Singular Series

3.1 Plain Euler products

The Bateman–Horn singular series for k -constellations is

$$\mathfrak{S}_k = \prod_p \sigma_k(p), \quad \sigma_k(p) = \frac{1 - \omega_k(p)/p}{(1 - 1/p)^k}. \quad (6)$$

With (5), $\ln \sigma_k(p) \approx (k - \omega_k(p))/p$, which is $+k/p$ at the 45 out of 46 non-resonant primes and $\approx -45k/p$ at each resonant prime. The two contributions cancel *on average*, so the product converges, but only conditionally: between consecutive resonant primes the partial product climbs by a factor $\approx (\ln p'/\ln p)^k$ and then drops by $1 - 46k/p$. Table 2 and Figure 2 show this saw-tooth. Even at $B = 10^4$ the partial product is still 2–3% below its limit, and at $B = 10^5$ it is 0.7–1.0% above it; a truncated product cannot certify its own convergence.

Table 2: Plain partial singular series $\mathfrak{S}_k(B) = \prod_{p \leq B} \sigma_k(p)$, computed with the exact root counts (script `compute_singular_series_v2.py`; the $B = 10^4$ row is also produced by the Version 1 script). Last row: exact values from Section 3.2.

Sieve bound B	$\mathfrak{S}_4(B)$	$\mathfrak{S}_5(B)$	$\mathfrak{S}_6(B)$	Note
100	4,772	39,661	329,634	no resonant prime yet
282	10,780	109,847	1,119,303	just before the first cliff
283	6,066	53,336	480,764	first cliff ($\times 0.563, 0.486, 0.430$)
1,000	7,947	74,537	710,658	partial recovery
5,000	6,668	60,315	555,276	
10^4	6,392	57,173	520,348	Version-1 truncation point
10^5	6,559	59,045	540,940	
10^6	6,502	58,410	533,966	
Exact	6,514	58,550	535,502	L -function evaluation, Eq. (7)

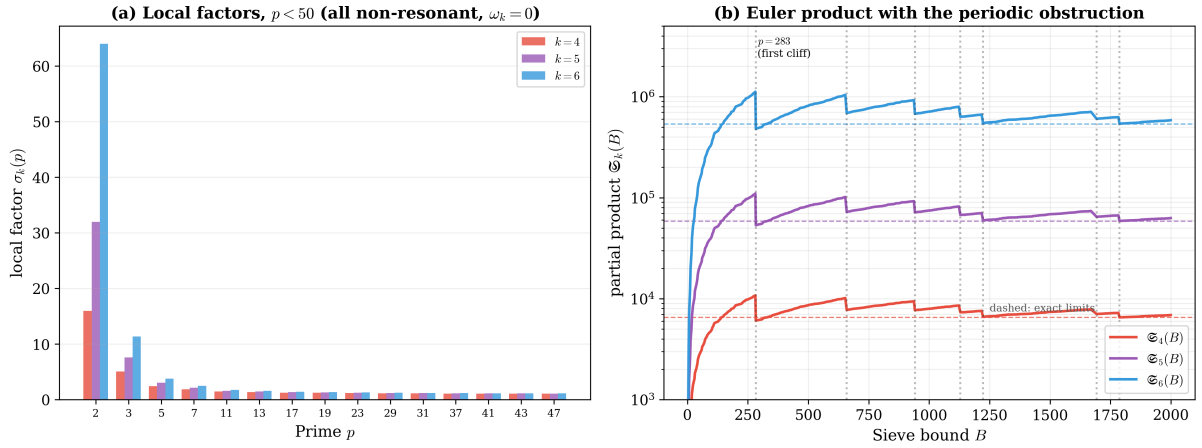


Figure 1: (a) Local factors for the non-resonant primes $p < 50$: $\sigma_k(p) = (p/(p-1))^k$. (b) Partial products $\mathfrak{S}_k(B)$ for $B \leq 2000$, showing the cliff at $p = 283$ and the subsequent saw-tooth; dashed lines are the exact values.

3.2 Exact evaluation via Dirichlet L -functions

Let χ run over the 45 non-principal Dirichlet characters modulo 47 (all primitive) and let $f = f(p)$ denote the order of p modulo 47, so that $\prod_{\chi \neq \chi_0} (1 - \chi(p)/p) = (1 - p^{-f})^{46/f} / (1 - 1/p)$ for $p \neq 47$. Multiplying and dividing (6) by $\prod_{\chi \neq \chi_0} L(1, \chi)^k$ gives

$$\mathfrak{S}_k = \left[\prod_{\chi \neq \chi_0} L(1, \chi) \right]^{-k} \prod_p B_k(p), \quad (7)$$

$$B_k(p) = \left(1 - \frac{\omega_k(p)}{p} \right) (1 - p^{-f})^{-46k/f} \quad (p \neq 47), \quad B_k(47) = \left(\frac{47}{46} \right)^k.$$

For a generic resonant prime ($f = 1$, $\omega_k = 46k$) one has $B_k(p) = (1 - 46k/p)(1 - 1/p)^{-46k} = 1 + O(k^2/p^2)$, and for a non-resonant prime ($f \geq 2$) $B_k(p) = 1 + O(k/p^2)$. The product in (7) is therefore absolutely convergent, with tail $O(k^2/(B \ln B))$ beyond a cut-off B . The L -values are computed exactly from the finite formulas for primitive characters [4, Thm. 4.9]; the quadratic

character gives the check $L(1, \chi_{47}) = \pi h(-47)/\sqrt{47} = 2.29124\dots$ with $h(-47) = 5$, and

$$\prod_{\chi \neq \chi_0 \pmod{47}} L(1, \chi) = 0.115169277\dots, \quad (8)$$

which by the analytic class number formula is the residue of the Dedekind zeta function of $\mathbb{Q}(\zeta_{47})$ at $s = 1$.

Table 3: Convergence of the accelerated product (7) with the cut-off B .

B	$\mathfrak{S}_1$	$\mathfrak{S}_2$	$\mathfrak{S}_3$	$\mathfrak{S}_4$	$\mathfrak{S}_5$	$\mathfrak{S}_6$
10^3	8.7146	78.341	724.56	6,535.9	58,377	530,038
10^4	8.6846	77.990	722.74	6,527.4	58,688	536,938
10^5	8.6829	77.945	721.92	6,514.7	58,552	535,529
10^6	8.6827	77.943	721.90	6,514.5	58,550	535,502
3×10^6	8.6827	77.943	721.89	6,514.4	58,549	535,495

We adopt

$$\mathfrak{S}_1 = 8.683, \quad \mathfrak{S}_2 = 77.94, \quad \mathfrak{S}_3 = 721.9, \quad \mathfrak{S}_4 = 6,514, \quad \mathfrak{S}_5 = 58,550, \quad \mathfrak{S}_6 = 535,500, \quad (9)$$

accurate to about 10^{-4} relative, and the inter-level ratios

$$\frac{\mathfrak{S}_2}{\mathfrak{S}_1} = 8.98, \quad \frac{\mathfrak{S}_3}{\mathfrak{S}_2} = 9.26, \quad \frac{\mathfrak{S}_4}{\mathfrak{S}_3} = 9.02, \quad \frac{\mathfrak{S}_5}{\mathfrak{S}_4} = 8.99, \quad \frac{\mathfrak{S}_6}{\mathfrak{S}_5} = 9.15, \quad \frac{\mathfrak{S}_6}{\mathfrak{S}_4} = 82.2. \quad (10)$$

The ratios are all close to 9, the value $\prod_{\chi \neq \chi_0} L(1, \chi)^{-1} \approx 8.68$ corrected by the overlap of shifted root sets at the small resonant primes. (Version 1 compared these ratios with “root-free” values; since the root-free product $\prod_{p \leq B} (1 - 1/p)^{-k}$ diverges like $(e^\gamma \ln B)^k$, that comparison has no limit and is withdrawn.)

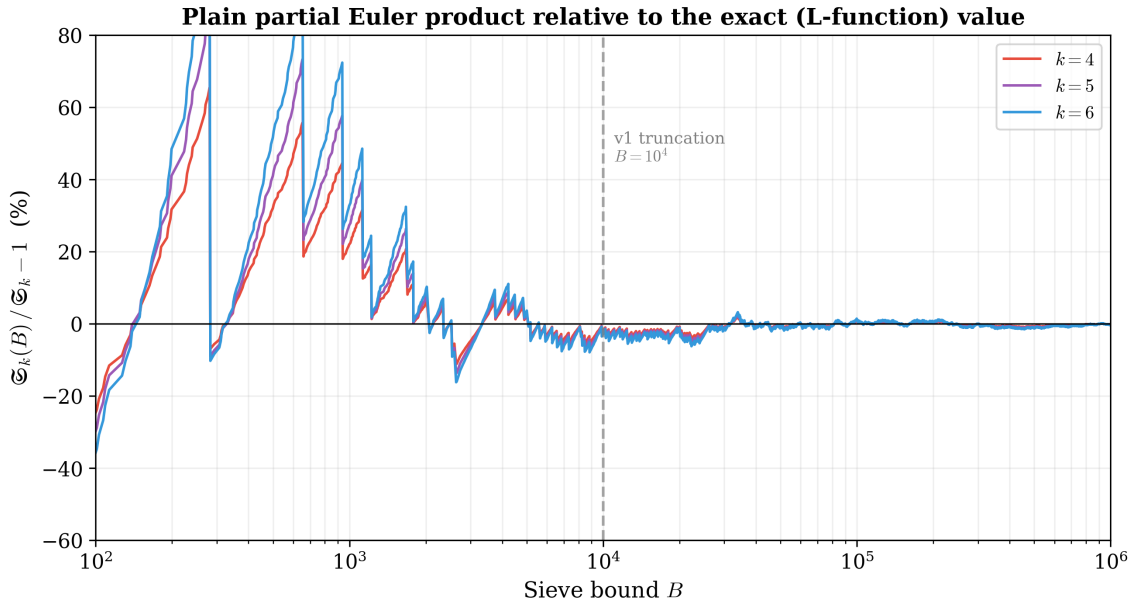


Figure 2: Relative deviation of the plain partial product $\mathfrak{S}_k(B)$ from the exact value, for $k = 4, 5, 6$ and $B \leq 10^6$. The oscillation is $\pm 20\%$ at $B \sim 10^3$ and still $\pm 1\%$ at $B \sim 10^5$; the Version-1 truncation at $B = 10^4$ happened to fall within 3% of the limit.

3.3 Parameter-free validation across the hierarchy

With (9) and (1), Bateman–Horn gives predictions without any fitted quantity. Table 4 compares them with the two censuses, converting maximal-run counts to position counts as explained in Section 1.1.¹

Table 4: Bateman–Horn predictions with the exact singular series versus the observed position counts. Deviations are in units of the Poisson standard deviation $\sqrt{E}$.

N	k	Census (as reported)	Positions	$E[C_k(N)]$	Deviation
2×10^9	1	18,121,562	18,473,571	18,464,404	+0.05% (+2.1 σ)
2×10^9	2	173,351	176,894	176,778	+0.07% (+0.3 σ)
2×10^9	3	1,749	1,779	1,752	+1.5% (+0.6 σ)
2×10^9	4	15	15	17.0	−0.5 σ
2×10^{11}	4	742	749	746.9	+0.3% (+0.1 σ)
2×10^{11}	5	7	7	5.88	+0.5 σ
2×10^{11}	6	0	0	0.047	−0.2 σ

The agreement at $k = 1$ and $k = 2$ —five and three significant figures respectively—is the strongest evidence that the singular series (9) and the integral (1) are correct in detail. The empirical constant $749/I_4(2 \times 10^{11}) = 6,533$ agrees with the exact $\mathfrak{S}_4 = 6,514$ to 0.3%, well inside the 3.7% Poisson uncertainty of a count of 749; Version 1’s calibrated value 6,385 was lower only because it used $(46 \ln t)^4$ and the run count 742.

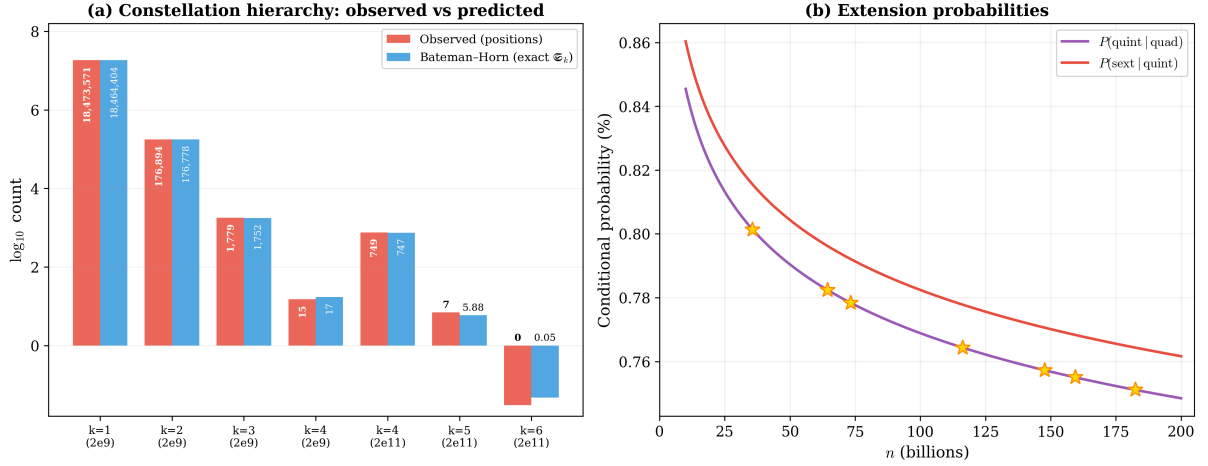


Figure 3: (a) Observed position counts versus the parameter-free Bateman–Horn prediction for $k = 1, \dots, 6$ (log scale; the first four bars refer to the pioneer zone, the last three to the full range). (b) Conditional extension probabilities $P(\text{quint} | \text{quad}) = (\mathfrak{S}_5/\mathfrak{S}_4)/\ln Q(n+4)$ and $P(\text{sext} | \text{quint})$; stars mark the 7 quintuplets.

¹The pioneer-zone categories of Part I are read as runs of exact length (“solitary” = isolated prime, “pair” = exactly two consecutive primes, etc.). Under this reading the total number of primes is $18,121,562 + 2 \times 173,351 + 3 \times 1,749 + 4 \times 15 = 18,473,571$. If instead “solitary” meant all primes, the deviation would be −1.9% (80 standard deviations), so the data themselves fix the convention.

4 The Seven Quintuplets

4.1 Catalog

Table 5 lists all 7 prime quintuplets. The digit counts are the exact lengths of $Q(n)$ (they are one larger than in Version 1 and in the Part II catalogue, which reported $\lfloor \log_{10} Q(n) \rfloor$ instead of $\lfloor \log_{10} Q(n) \rfloor + 1$); within each quintuplet all five values have the same number of digits.

Table 5: All 7 prime quintuplets for $Q(n)$ in $n \leq 2 \times 10^{11}$. “Rank” is the position in the sorted list of 742 quadruplets.

#	Starting n	Digits of Q	$n/10^{11}$	Quad rank	Gap from previous
1	35 676 017 721	488	0.357	#154	—
2	64 482 563 907	499	0.645	#251	28.8 B
3	73 292 417 435	502	0.733	#284	8.8 B
4	116 255 850 744	511	1.163	#444	43.0 B
5	147 743 683 226	516	1.477	#570	31.5 B
6	159 430 471 996	517	1.594	#611	11.7 B
7	182 501 065 420	520	1.825	#681	23.1 B

4.2 Spatial Distribution

The quintuplets span the full search range: one in $[0, 50\text{B})$, two each in $[50, 100)$, $[100, 150)$ and $[150, 200)$ billion (Figure 4). The six gaps range from 8.8 to 43.0 billion (mean 24.5 B, sample standard deviation 12.8 B). A Kolmogorov–Smirnov test of the seven positions against the Bateman–Horn cumulative distribution (proportional to $I_5(n)$) gives $D = 0.265$, $p = 0.62$; against a uniform distribution on $[0, 2 \times 10^{11}]$ it gives $D = 0.180$, $p = 0.95$ (the value quoted in Version 1 corresponds to the latter). With seven points neither test has power to detect anything but gross clustering; the data are simply consistent with the model.

5 The Conditional Extension Model

Given a quadruplet starting at n , the conditional probability that it extends to a quintuplet is

$$P(\text{quint} \mid \text{quad at } n) \approx \frac{\mathfrak{S}_5}{\mathfrak{S}_4} \cdot \frac{1}{\ln Q(n+4)} = \frac{8.99}{46 \ln n + \ln 47}. \quad (11)$$

Summing (11) over the 742 catalogued quadruplets ($\langle \ln n \rangle = 24.90$) gives

$$E_{\text{cond}}[C_5] = \sum_{i=1}^{742} \frac{8.99}{\ln Q(n_i+4)} = 5.82, \quad (12)$$

to be compared with the direct integral $E[C_5] = \mathfrak{S}_5 I_5 = 5.88$ (the small difference reflects the Poisson scatter of the quadruplet positions themselves). The observed 7 lies inside the 1σ interval $[3.5, 8.3]$ of a Poisson variable with mean 5.88, and $P(C_5 \geq 7 \mid 5.88) = 0.37$. Under naïve independence ($\mathfrak{S}_5/\mathfrak{S}_4 = 1$) one would expect only 0.65 quintuplets; the factor of 9 is entirely the singular-series ratio, i.e. the fact that a quadruplet has already “paid” the sieve cost of the resonant primes for four of the five positions.

6 The Sextuplet Boundary

At $N = 2 \times 10^{11}$,

$$E[C_6] = \mathfrak{S}_6 I_6(2 \times 10^{11}) = 535,500 \times 8.80 \times 10^{-8} = 0.047, \quad (13)$$

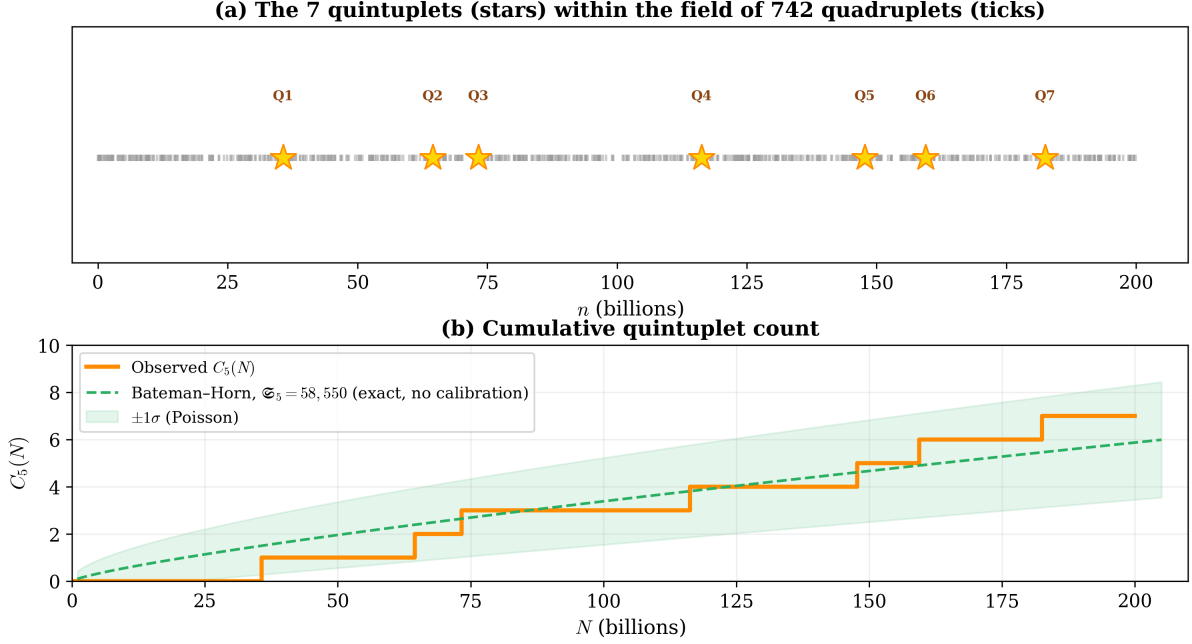


Figure 4: Top: quintuplet positions (★) within the 742-quadruplet field. Bottom: cumulative count $C_5(N)$ versus the parameter-free Bateman–Horn curve with $\mathfrak{S}_5 = 58,550$ and its Poisson $\pm 1\sigma$ band.

so the probability of at least one sextuplet in the present range is $1 - e^{-0.047} = 4.6\%$: its absence is the expected outcome. Solving $E[C_6(N^*)] = 1$ gives

$$N^* \approx 1.03 \times 10^{13} \quad (14)$$

(1.01×10^{13} with the $(46 \ln t)^6$ convention of Version 1), a 52-fold extension of the current search, where $Q(n)$ has 601 digits. Table 6 and Figure 5 give the outlook.

Table 6: Predicted counts (exact singular series, Eq. (1)).

Search bound N	$E[C_5]$	$E[C_6]$	$P(C_6 \geq 1)$	Status
2×10^{11} (current)	5.9	0.047	4.6%	7 / 0 observed
10^{12}	21	0.16	15%	
2×10^{12}	38	0.28	24%	
5×10^{12}	80	0.57	43%	coin flip
10^{13}	141	0.98	62%	
1.03×10^{13}	145	1.00	63%	sextuplet boundary
2×10^{13}	251	1.69	82%	
5×10^{13}	538	3.52	97%	
10^{14}	961	6.15	99.8%	sextuplets routine

7 The Suppression Factor

The ratio of successive position counts is, by (1),

$$R_{k \rightarrow k+1}(N) = \frac{E[C_k(N)]}{E[C_{k+1}(N)]} = \frac{I_k(N)}{I_{k+1}(N)} \bigg/ \frac{\mathfrak{S}_{k+1}}{\mathfrak{S}_k} \approx \frac{46 \ln N_{\text{eff}} + \ln 47}{9.0}, \quad (15)$$

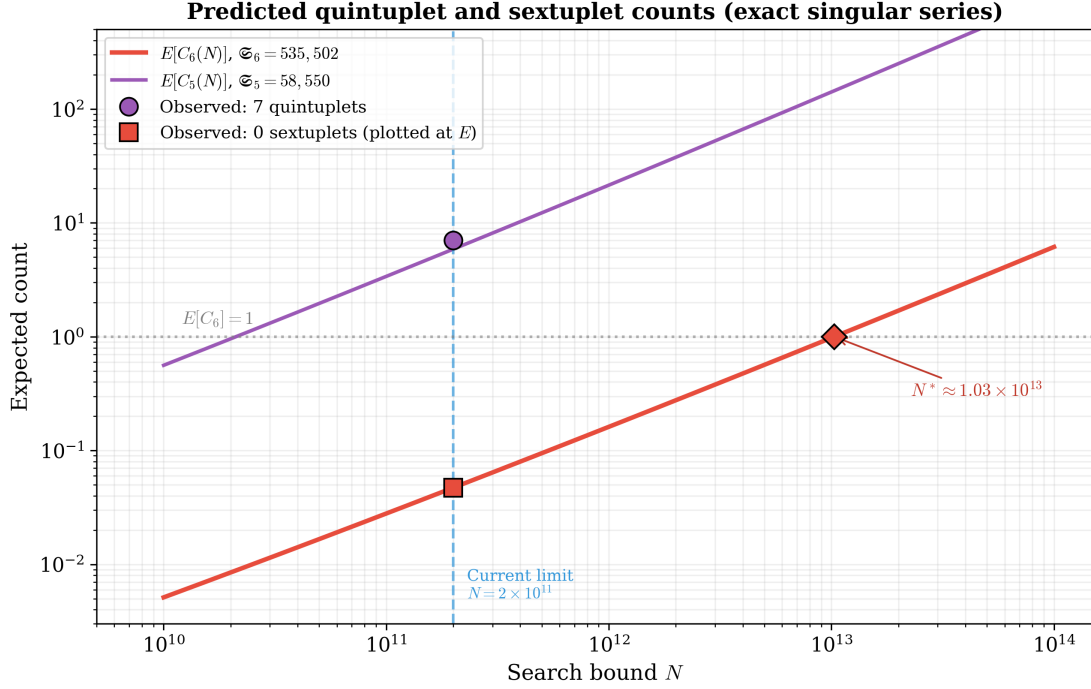


Figure 5: $E[C_5(N)]$ and $E[C_6(N)]$ versus search bound. The diamond marks the sextuplet boundary.

where N_{eff} is a logarithmic mean of the range. The factor is therefore *not* a universal constant (as Version 1 claimed) but grows logarithmically with the search bound: Table 7.

Table 7: Inter-level suppression factors: Bateman–Horn prediction (15) versus observed ratios of position counts.

Range	Transition	Predicted R	Observed R
$N = 2 \times 10^9$	$1 \rightarrow 2$	104.4	104.4
	$2 \rightarrow 3$	100.9	99.4
	$3 \rightarrow 4$	103.1	118.6
	$4 \rightarrow 5$	103.0	—
$N = 2 \times 10^{11}$	$3 \rightarrow 4$	126.9	—
	$4 \rightarrow 5$	127.1	107.0
	$5 \rightarrow 6$	124.7	—

The pioneer-zone ratios cluster around 103 and the full-range ratios around 126, exactly as (15) predicts from $\ln(2 \times 10^{11})/\ln(2 \times 10^9)$; the two observed outliers (118.6 and 107.0) involve counts of 15 and 7 and are within Poisson noise. The near-equality of the ratios $\mathfrak{S}_{k+1}/\mathfrak{S}_k \approx 9$ at every level is what makes the factor almost independent of k at fixed N .

8 Discussion

8.1 The Bifurcated Sieve

The periodic obstruction at $p \equiv 1 \pmod{47}$ is a property of the cyclotomic structure, not of the exponent 47 alone: for $D_q(n) = n^q - (n-1)^q = \Phi_q(n, n-1)$ with any prime q , every prime factor of every value is $\equiv 1 \pmod{q}$, $\omega_1(p) = q - 1$ for $p \equiv 1 \pmod{q}$ and zero otherwise, and the

singular series is given by (7) with the L -values modulo q . Larger q gives a longer unobstructed corridor (no root below the first prime $\equiv 1 \pmod{q}$) and hence larger constants, but also a smaller density $1/(q-1)$ of resonant primes with individually more destructive cliffs.

8.2 Quintuplets as a Bateman–Horn Prediction

The 7 observed quintuplets match the parameter-free prediction $E = 5.88$ within 0.5σ , and the conditional model gives 5.82. Together with the 0.05% and 0.07% agreements for primes and pairs in the pioneer zone (Table 4), this is a precise validation of Bateman–Horn for a system of six polynomials of degree 46 at ~ 500 -digit scale.

8.3 Sextuplet Outlook

Part II reports ~ 240 CPU-core-hours for the range 2×10^{11} [6]. Reaching $N^* \approx 1.03 \times 10^{13}$ multiplies the range by 52 and the per-candidate cost by roughly $(601/500)^{1.6} \approx 1.3$ for 600-digit modular arithmetic, i.e. $\sim 16,000$ core-hours or about four weeks on a 26-core workstation—feasible. (Version 1’s “12,500 core-hours, 2 weeks” omitted the digit-size factor and misconverted the hours.) Note that at N^* the *expected* number of sextuplets is one; the probability of actually having found one is 63% (50% is reached at $N \approx 6.5 \times 10^{12}$), and reaching 90% requires $N \approx 2.9 \times 10^{13}$.

9 Conclusion

$Q(n) = n^{47} - (n-1)^{47}$ is the homogenised cyclotomic polynomial $\Phi_{47}(n, n-1)$: all its prime factors are $\equiv 1 \pmod{47}$, it has no roots modulo non-resonant primes and 46 roots modulo every resonant prime. This bifurcated structure makes the Bateman–Horn Euler product oscillate strongly, but factoring out $\prod_{\chi \neq \chi_0} L(1, \chi) = 0.11517$ yields exact singular series $\mathfrak{S}_4 = 6,514$, $\mathfrak{S}_5 = 58,550$, $\mathfrak{S}_6 = 535,500$. Without any calibration these reproduce the whole observed hierarchy, from 18.47 million primes in the pioneer zone (0.05%) to 749 quadruplet positions (0.3%), 7 quintuplets ($E = 5.9$) and 0 sextuplets ($E = 0.047$) in the full range. The first sextuplet is expected near $N^* \approx 1.0 \times 10^{13}$, and the inter-level suppression factor grows as $\approx 46 \ln N/9$, from ≈ 103 in the pioneer zone to ≈ 127 at the present frontier.

Data and Code Availability

Complete data and code for this paper:

<https://github.com/Ruqing1963/Q47-Quintuplet-Sextuplet-Boundary>

The Version 2 material consists of:

- `compute_singular_series_v2.py` — plain and L -function evaluation of $\mathfrak{S}_k$;
- `predictions_v2.py` — all predictions, tests and tables;
- `generate_figures_v2.py` — Figures 1–5;
- the corresponding `data/*_v2.csv` files and `CHANGELOG_v2.md`.

Quadruplet census data (Part II):

<https://github.com/Ruqing1963/Q47-Deep-Space-Quadruplet-Census>

Zenodo archives:

Part III, this paper, Version 2: <https://doi.org/10.5281/zenodo.23178170>

Part III, Version 1: <https://doi.org/10.5281/zenodo.18728917>

Part II: <https://zenodo.org/records/18728540>

Part I: <https://zenodo.org/records/18701355>

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